

1. Administrative & Study Identification

Full Study Title: HU-F-AIM - A Prospective, Interventional Study to Evaluate HU-resistance in Polycythemia Vera Patients Who Meet Predictive Parameters Identified in the Machine Learning Project PV-AIM **Short Title/Acronym:** HU-F-AIM **Protocol Number:** CINC424BDE15 **Other Identifier:** 2022-502338-20-00 (EU CTIS number) **NCT Number:** NCT05853458 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Hydroxyurea (HU) **Phase of Development:** Phase IV **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate and validate the proportion of patients with hydroxyurea (HU) resistance or intolerance within 6–9 months after de novo HU treatment initiation in the presence of specific PV-AIM HU-RES predictors. **Secondary Objectives:** To evaluate patient-reported outcomes (using the MPN-SAF TSS score) and the general safety and tolerability of HU treatment in this targeted patient population.

2.2 Background & Rationale Polycythemia vera (PV) is a myeloproliferative neoplasm associated with increased thromboembolic (TE) risk and hematologic complications. While Hydroxyurea (HU) is the most common first-line cytoreductive therapy, resistance to HU (HU-RES) develops in a significant subset of patients. The machine learning-based PV-AIM project identified specific predictive markers for HU failure: Hemoglobin (HGB) < 15.5 g/dL (9.62 mmol/L) and Red Cell Distribution Width (RDW) ≥ 17%. This study aims to prospectively validate these predictors in a real-world clinical setting, facilitating earlier intervention and optimization of treatment strategies.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Intervention Model:** Single Group Assignment

3.2 Enrollment, Status & Duration **Overall Status:** Terminated **Target \$N\$:** 300 adult patients **Number of Sites:** Approximately 30 to 40 sites in Germany, with potential extension to other countries if necessary to achieve the target population. **Actual Study Start Date:** 28-Jul-2023 **Actual Primary Completion Date:** 26-Feb-2026 **Actual Study Completion Date:** 26-Feb-2026 **Maximum Treatment Duration:** Up to 15 months per participant

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult PV patients (Ages 18-99 years). 2. High-risk (age ≥ 60 years and/or prior history of thrombosis); OR Low-risk showing signs of disease progression (myeloproliferation) such as: increase in spleen size or symptomatic splenomegaly, platelet increase to $> 1,000,000/\mu\text{L}$, WBC increase to $> 15,000/\mu\text{L}$, frequent (> 10 per year) or increasing frequency of phlebotomies. 3. Increasing risk of thromboembolism and bleeding: new thromboembolism and/or hemorrhagic complications, microcirculation disorders despite acetyl salicylic acid (ASA) $2 \times 100 \text{ mg/day}$, restricted feasibility or intolerance of phlebotomies, symptomatic iron deficiency, uncontrolled increase in hematocrit, or severe/distressing disease-related symptoms. 4. Meeting PV-AIM predictors for HU failure at the start of treatment: $\text{HGB} < 15.5 \text{ g/dL}$ (9.62 mmol/L) and $\text{RDW} \geq 17\%$. 5. Female participants of childbearing potential must have a negative serum pregnancy test within 72 hours prior to receiving the first dose of study treatment.

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women; sexually active males unwilling to use a condom during intercourse while taking study treatment and for at least 3 months after stopping treatment. 2. HIV patients treated with nucleoside reverse transcriptase inhibitors like didanosine and stavudine. 3. Patients with post-polycythemia vera myelofibrosis (post-PV MF) or accelerated phase/blast phase myeloproliferative neoplasm acute myeloid leukemia (AP/BP-MPN AML). 4. Patients with a contraindication to HU according to the SmPC (severe bone marrow depression, leukopenia $< 2.5 \times 10^9/\text{L}$, thrombocytopenia $< 100 \times 10^9/\text{L}$, severe anemia $< 10 \text{ g/dL}$ HGB). 5. Patients with rare hereditary galactose intolerance, total lactase deficiency, or glucose-galactose malabsorption in their past medical history. 6. Active uncontrolled infection that is considered by the Investigator as a reason for exclusion.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: De novo Hydroxyurea (HU) treatment standard dosing, adjusted per clinical guidelines. **Route of Administration:** Oral (Capsules) **Duration of Treatment:** Up to 15 months (observation period for HU-resistance/intolerance).

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant treatments to manage disease progression such as acetyl salicylic acid (ASA) and phlebotomies. **Prohibited:** Nucleoside reverse transcriptase inhibitors like didanosine and stavudine for HIV patients.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -14 to 0 | Informed Consent, Medical History, Baseline Predictor Labs (HGB, RDW), Pregnancy Test (within 72 hours of dose) | | **Baseline** | Day 1 | Treatment Initiation | | **Interim** | Weeks 2 to 12 (Q2W) & Weeks 18 to 54 (Q6W) | Safety Labs, HU-resistance/intolerance assessment, Patient-Reported Outcomes | | **End of Study** | Week 60 | Final assessments (EOT), followed by 30-day safety follow-up and 3-month follow-up for resistant/intolerant patients |

(Note: Data extraction from screening up to week 60 mapped according to the HU-F-AIM study design)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of patients demonstrating HU resistance or intolerance within 6–9 months after de novo HU treatment initiation in the presence of the PV-AIM HU-RES predictors at baseline. **Measurement Method:** Modified European LeukemiaNet (ELN) criteria for HU resistance and intolerance, calculating the rate together with the respective 95% confidence interval (CI).

7.2 Secondary & Exploratory Endpoints 1. Patient-reported outcomes evaluated using the Myeloproliferative Neoplasm Symptom Assessment Form Total Symptom Score (MPN-SAF TSS). 2. Overall safety and tolerability of the HU treatment profile during the observation period.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during Q2W and Q6W clinic visits, extending to a 30-day safety follow-up post-treatment. **Serious Adverse Events (SAEs):** Reported according to standard phase IV pharmacovigilance regulations. **Data Monitoring Committee (DMC):** Supervised by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Patients with available resistance predictors at day 1 (or screening) and data available on HU-RES/intolerance at the End of Treatment (EOT). **Sample Size**

Justification: $N = 300$ allows for a precision in terms of the 95% Confidence Interval width of ± 0.072 , assuming ~40% of HU-naïve patients meet the PV-AIM criteria and ~80% of those show HU failure within 3-6 months. **Handling of Missing Data:** Patients with no available data on HU-RES/intolerance at the EOT will be

excluded from the primary endpoint analysis. No formal statistical testing; rates and 95% CIs evaluated.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite tracking in the initial 12 weeks followed by 6-week interval tracking. **Audit Trail:** Standard Phase IV eCRF data locking and query resolution managed by the sponsor.

11. Study Identifiers & Governance

Primary ID: CINC424BDE15 **Other ID:** 2022-502338-20-00 (EU CTIS number) **Registry Number:** NCT05853458 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** HU-F-AIM **Official Regulatory Title:** HU-F-AIM - A Prospective, Interventional Study to Evaluate HU-resistance in Polycythemia Vera Patients Who Meet Predictive Parameters Identified in the Machine Learning Project PV-AIM

12. Clinical Framework (Polycythemia Vera Specific)

Primary Condition: Polycythemia Vera (PV) **Diagnosis Threshold:** Baseline predictors: \$HGB < 15.5\$ g/dL (9.62 mmol/L) & \$RDW \ge 17\%\$ **Medical Methodology:** Cytoreductive Therapy (Hydroxyurea) **Optimization Target:** Early identification and management of HU resistance/intolerance

13. Study Procedures & Methodology

Management Pathway: Hydroxyurea (HU) initiation accompanied by proactive clinical monitoring **Education Protocol (HCP):** Recognition of PV-AIM predictive biomarkers to anticipate treatment failure **Education Protocol (Patient):** Onsite tracking of patient-reported outcomes via MPN-SAF TSS **Quality Audit Mechanism:** Real-world validation of machine learning algorithms utilizing modified ELN criteria

14. Observation & Follow-Up Schedule

Total Duration: 15 months (plus 30-day and 3-month follow-up windows) **Onsite Visit Cadence:** Baseline, Q2W (Weeks 2-12), Q6W (Weeks 18-60) **Remote Monitoring Frequency:** Inter-visit assessments as clinically appropriate **Checklist Review Interval:** Routine at every specified clinic visit

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					